

GLOBAL MODELS, INDIAN MOVES

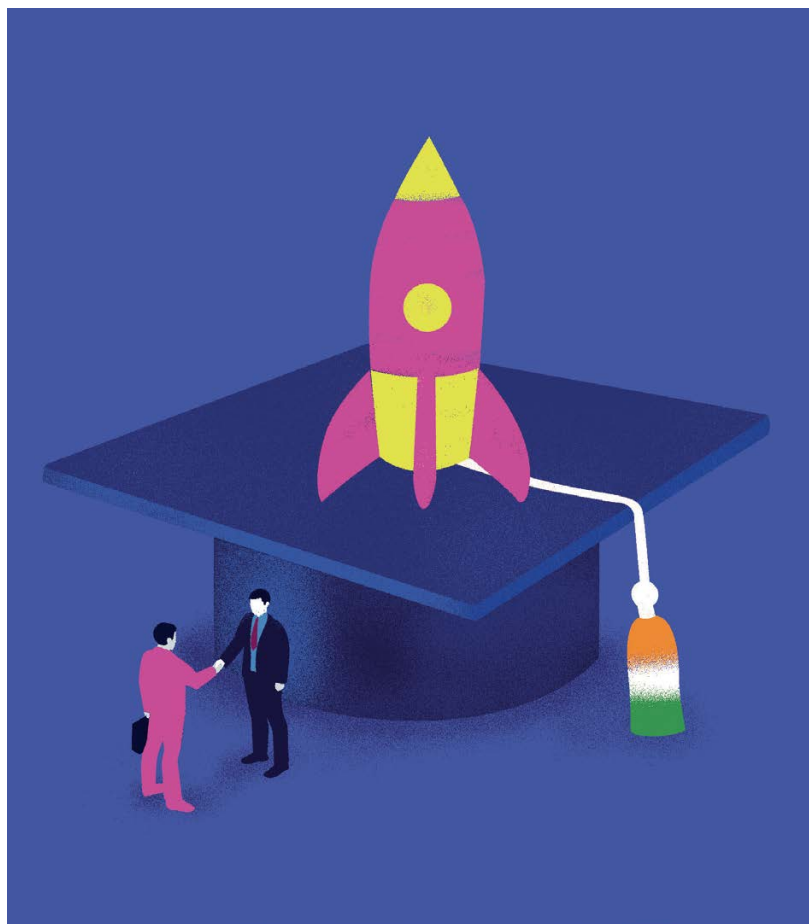
India can and is following established playbooks for aligning research, industry, and government.

The US bridged academia and industry with the 1980 Bayh-Dole Act. Germany's Fraunhofer Society commercialised research and earns €836 million annually. China rewards province-level tech transfers.

Domestic successes validate this approach. IIT Madras's ₹1.25-crore bet on Ather Energy is set for a 40x IPO return. Boosted by this, innovation continues to emerge, like Professor Chakravarthy's air taxis from IIT-Madras. Large industry is also investing: Ola has created a Battery Innovation Center and Tata Steel is creating next-gen alloys at its research centre. Corporates like Shell and Qualcomm now fund Indian deep-tech companies. With the private sector expected to fund 70% of the RDI scheme, this trend will accelerate.

India is also emulating the US government's role as early-stage funder and customer. ISRO is opening its launch pads to start-ups, and HAL and L&T are manufacturing launch vehicle components. The iDEX programme is funding defence start-ups, while the ₹76,000-crore Semiconductor Mission supported Micron's \$2.75-billion investment. The Government e-Marketplace now supports over 30,000 start-up suppliers. Sebi's Innovators Growth Platform is easing listing rules for IP-heavy firms.

The government-academia divide is also being bridged, following global models of open, collaborative funding and shared infrastructure. Further, the ₹50,000 crore NRF will bankroll university research, and BIRAC has already funded 500-plus biotech start-ups. On infrastruc-



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ture, the Bangalore Bioinnovation Centre offers subsidised wet labs and pilot scale lines, nurturing 100+ life science start-ups. To scale up these successes, the government is now establishing Science & Technology clusters in Bengaluru and Hyderabad.

THE WAY FORWARD

India must now ensure strong innovation pipelines by raising the share of R&D funding for higher education, distributed through competitive, milestone-based grants. To move innovations from lab to market, a Bayh-Dole-style law

would grant universities ownership of publicly funded IP to incentivise commercialisation, complemented by professional tech-transfer offices and regional science parks.

The government can also modernise its role by shifting procurement towards technical merit and start-up quotas to de-risk new technologies. A dedicated agency, modeled on ARPA, should be established with significant autonomy and high failure tolerance to make bold, strategic bets in fields like AI and biotech. **BT**

Views are personal